The Voice Bank Corpus: Design, Collection and Data Analysis of a Large Regional Accent Speech Database

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Abstract

The University of Edinburgh has started the development of a new speech database, the Voice Bank corpus, specifically designed for the creation of personalised synthetic voices for individuals with speech disorders. This corpus already constitutes the largest corpora of British English currently in existence, with more than 300 hours of recordings from approximately 500 healthy speakers. New recordings are continuously being made in order to get the best coverage of the different combinations of regional accents, social classes, age and gender across Britain. This paper describes the motivation and the processes involved in the design and recording of this corpus as well as some analysis of its content. The paper concludes with our future plans to further extend this corpus and to overcome its current limitations.

Index Terms: Corpus Design, Text Selection, Speech Synthesis, Voice Banking.

1. Introduction

Speech databases are a key component for corpus-based text-to-speech (TTS) systems, and therefore speech corpus design has becoming a very active research field during the last 15 years. A lot of work was initially focused on the coverage optimization of the large speech databases required by unit-selection TTS [1]. These databases typically contain several hours of recording from a single speaker. Although using such database can yield to high-quality speech synthesis, the cost associated with the creation of each voice is very high. However the emergence of statistical parametric speech synthesis [2] based on hidden Markov models (HMMs) has introduced a paradigm shift. With HMM-based speech synthesis, it is now possible to build a voice from a small amount of speech by adapting existing voice models pre-trained on many speakers [3]. This technique, referred to as speaker adaptation, has already been successfully applied to "non traditional TTS" corpora such as ASR corpora, which contains a large number of speakers [4]. This leads to the possibility of producing a wide variety of voices automatically and at much lesser cost. The Voice Bank corpus follows this new approach as it consists in a collection of relatively short recordings (about one hour each) from a large number of speakers. This corpus is a result of a collaborative project for voice banking and voice reconstruction initiated by the University of Edinburgh, the Eaun MacDonald Center for Motor Neuron Disease, and the Anne Rowling Regenerative Neurology Clinic. The voice reconstruction aims at recreating a computerized version of their voice for individuals with degenerative speech disorders [5]. The Voice Bank corpus plays a central role in the voice reconstruction process. It was designed to provide a large catalogue of healthy voices over which we can train a set of average voice models specifically matched to the voice characteristics of each patient. Therefore, the first specification of the Voice Bank corpus was to cover a wide variety of accents and voice identities across the UK, and with an initial focus on Scotland where the project started. The Voice Bank corpus already comprises more than 300 hours of speech data from approximately 500 healthy speakers, and the number of recorded speakers is increasing continuously. At this point, this corpus is already larger than the WSJCAM0 database [6], a standard UK speech research database. It is also larger than all existing corpus of British accents such as the ABI corpus [7]. The Voice Bank corpus already covers a wide coverage of regional accents across the UK. An illustration of the geographical distribution of the speakers recorded so far is shown in Figure 1.

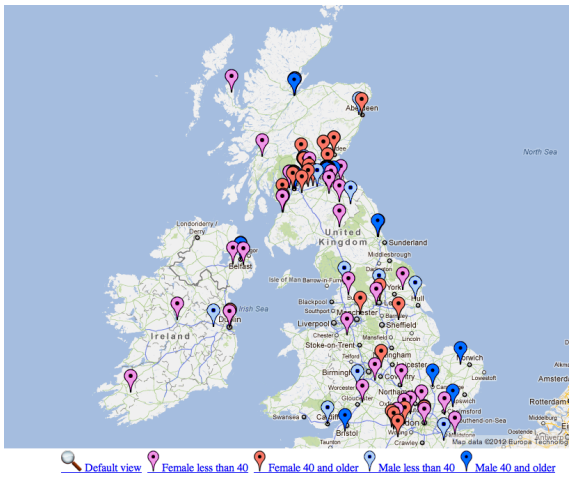


Figure 1: Geographical distribution of the speakers recorded for the Voice Bank corpus

This paper describes the procedures that were used to design and record the Voice Bank corpus. The section 2 addresses the prompt design with coverage optimization and readability constraint. The section 3 details the recruitment and recording of the speakers, as well as the metadata information collected from them. The section 4 presents some statistical analysis based on the metadata information. We finally conclude by discussing the current shortcomings of the Voice Bank corpus and presenting some ideas to improve its coverage and usability.

2. Prompt Design

This section details the methodology that we followed to create the text scripts used for the recording of the Voice Bank corpus.

2.1. Text materials

The text materials used to generate the scripts were collected from 3 sources:

- 3000 articles of the Scottish Herald newspaper
- The Rainbow passage
- Accent elicitation passage from the Speech Accent Archive

The Voice Bank speech corpus was initially designed with a focus on Scottish accent as its first use was for the voice reconstruction of Scottish patients. Therefore the Scottish Herald was chosen as the main text resource because it comprises a number of Scottish proper nouns such as cities names. The articles of the Scottish Herald are representative of different topics (arts, politic, social issues, sports) and styles (narrative, interview). These articles were converted from their original HTML format in a unique sentence-based textual format and cleared form any malformed symbol. They represent 1300000 sentences in total and constitute the principal source for the script selection. The Rainbow passage is a set of 20 sentences covering a wide variety of consonants clusters. These sentences are commonly used by speech therapists for the assessment of articulation disorders. We decided to systematically include the Rainbow passage in all scripts in order to follow the same script design for the recording of healthy donors and patients. The accent elicitation passage is also included systematically at the beginning of the scripts. This passage corresponds to a set of 5 sentences used by the Speech Accent Archive to characterize the different English accents. It contains most of the consonants, vowels and clusters of Standard English [8].

2.2. Coverage optimization

Each speaker was asked to read out a script of 425 sentences, which represents just about one hour of recording. The first 25 sentences correspond to the Rainbow passage and the Accent elicitation passage. The remaining 400 sentences are selected from the Scottish Herald text corpus using a greedy algorithm in order to maximise the coverage at both phonetic and prosodic levels. In order to define a coverage measurement at the phonetic level, the text corpus is first phonetized using Festival [9] and the Combilex lexicon [10]. Then, the triphone frequency distribution is estimated using classical Good-Turing smoothing technique [11]. Finally, the sentence selection is performed iteratively by selecting the sentences that:

- Maximize the increase of the triphone coverage, defined as the cumulative frequency of all the triphone units collected from the text corpus.
- Balance the distributions of number of syllables and number of phrases per sentence.

These two criteria are equivalent to the selection of the “most frequent unit first” at the phonetic level and the “less frequent unit first” at the prosodic level. In order to have a sufficient number of representatives for each triphone, a minimum of 15 instances per triphone is required to consider that this triphone is fully covered.

On the contrary to corpora designed for speaker-dependent speech synthesis such as CMU-ARTIC [12], the Voice Bank corpus is designed for speaker-independent/adaptive speech synthesis where the data from multiple speakers is pooled together in order to train average voice models. Therefore the coverage optimisation presented above is performed over several speakers and the resulting set of selected sentences is split into smaller sets of 400 sentences for each speaker. In this way, each speaker records a different script and the total phonetic coverage can be improved.

2.3. Readability pre-filtering

One downside of text selection methods based only on maximal coverage criteria is that some sentences with complex words or incorrect grammar are often selected. When reading such sentences, the speakers could make mistakes, and the recordings often sound unnatural. The meaningless sentences also do not help in prosody training of HMM-based speech synthesis. In order to address this problem, we have designed a pre-selection filter based on a readability measure that retains only easy-to-read sentences from the text corpus. The coverage optimization presented above is applied only on this pre-selected subset of sentences. Earlier research on text readability has established a relationship between readability and very simple text properties (e.g. words per sentence, average number of syllables per word) by multiple correlation analysis of human graded text [13]. More sophisticated readability measures have since been based on language modeling techniques in order to assess both lexical and syntactic complexity [14] as well as grammatical complexity [15]. The readability measure designed’ for the Voice Bank corpus uses a mix of all these measures, and the following criteria are used to filter out overly complex sentences:

1) Sentences with words that are not found in the Combilex lexicon are excluded.
2) The number of words per sentence is limited between lower and upper bounds. Long sentences are normally more difficult to read and it is not efficient to record very short sentences.
3) The number of syllable per word is limited by an upper bound.
4) Sentences with less frequent syllable bigrams are excluded.
5) Sentence that contains noun phrases longer than a given upper bound are excluded.
6) Sentences with less frequent part-of-speech (POS) trigrams are excluded, as these sentences are more likely to be ungrammatical.

All the language statistics are learnt from the Scottish Herald corpus. Several tests recordings were made to adjust the threshold values and assess the readability of the selected sentences. Finally, the readability filter reduces the text corpus down to 450000 sentences and the coverage optimization is run over this subset in order to generate the text scripts. The readability constraint reduces the coverage rate achievable for a given number of sentences as shown on Figure 2. However this is not an issue here since the coverage optimization is performed across several speakers and less than 3500 sentences (8 speakers) are enough to reach full trigram coverage.

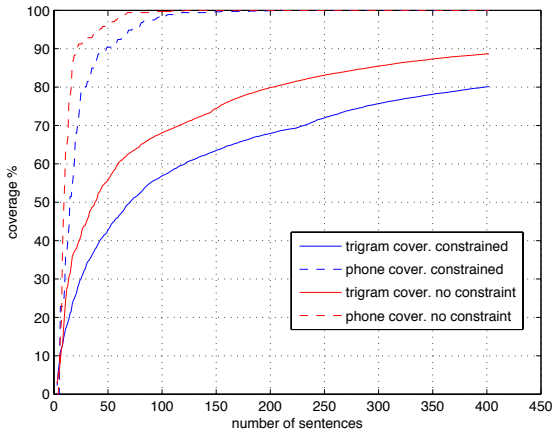


Figure 2: Phonetic coverage rate per number of selected sentences, with and without readability constraint.

3. Data Collection

This section details the procedures that we followed to recruit and to record the speakers, as well as the additional information collected from them.

3.1. Speakers recruitment

Speakers were recruited by advertising throughout University of Edinburgh, local newspapers and carers associations. This advertising campaign was limited to Edinburgh and around. However, effort was made to take advantage of the University's diverse population in order to cover a wide range of regional accents across the UK. The advertisement asked for native speakers of British English who would like to contribute to the Voice Bank project by reading out a text script for an hour. The recruitment was based on free volunteering only and the speakers had to sign a consent form detailing the conditions of use of their recordings and the confidentiality of their personal information.

3.2. Recording conditions

The Voice Bank corpus was recorded in a semi-anechoic chamber at the University of Edinburgh. We used two high quality microphones: a Sennheiser MKH 800 (omnidirectional diaphragm condenser with very wide bandwidth) and a DPA 4035 (headset-mounted condenser). All recordings were made at 96kHz sampling frequency and 24 bits per sample using ProTools HD hardware. The recordings are then downsampled at 48kHz. This technique is known as over-sampling method for noise reduction. Some additional recordings were made in the Scottish parliament in order to get a better coverage of the various regional accents in Scotland. These recordings were made in a sound booth using an AKG-C414 microphone and a Focusrite Forte USB audio interface connected to a laptop. An assistant monitored all the recordings, and the speakers were asked to repeat any mispronounced sentence.

3.3. Speakers categorisation

Appropriate information about the speaker characteristics is a fundamental component of a speech corpus since it can help to organise the speech data. The meta-data collected for the Voice Bank corpus consists of the speaker's age, gender, occupation, regional accent and childhood area, as these factors are likely to contribute to the characteristics of the voice. However, one difficulty here is the definition of what constitutes a regional accent. There are some accent variations that contain some acoustic parameters of an overall accent but are very distinctive to locals, who may claim that they are completely different accents. On the contrary, some speakers may have spent years in very different regions and as a result have a mix of different accents. Therefore, the speakers were asked two types of information about their accent:

1) The closest match from a list of all the counties in England, Wales, Scotland and Northern Ireland.
2) An additional commentary to describe their accent.

British accents are usually not defined at the county level, however this gives a good sampling of the possible local variations and can be converted into broader accent classification such as the accents list defined in [16] or even larger geographical regions (Southern and Northern England, Wales, Scotland, Northern Ireland). It is also well-known that the accent is strongly dependent of the speaker's social class [17]. Therefore, the speakers' occupation is used to derive their social class according to the NS-SEC classification standard [18]. NS-SEC is the primary social classification used in the United Kingdom for all official statistics and surveys. It has 8 main categories (known as the 'analytical' scale), but a shorter version with 5 classes can be used for small-scale analyses. We are using here a variant of the 5 classes version with 2 additional classes for students and unemployed:

A) Managerial and administrative occupations
B) Intermediate occupations
C) Small employers and own account workers
D) Lower supervisory and technical occupations
E) Semi-routine and routine occupations
F) Full-time students
G) Unemployed

The age, gender, accent and social class descriptors can serve as predictors of the perceived similarity between speakers [19]. These predictors are, for instance, used to pool together multiple speakers in order to create specific average voice models and to select the best voice donors for any given patient.

4. Corpus Analysis

The Voice Bank corpus should ideally cover a wide variety of British accents and other voice characteristics. We analyze here the distributions of speakers in the Voice Bank corpus, according to their gender, age, accent and social class. Theses statistical analyses could be used in the selection of new speakers in order to balance the corpus coverage. The repartition of speakers according to their gender and age is shown in Figure 3.

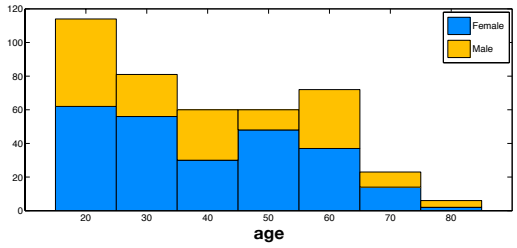


Figure 3: Distribution of speakers according to their age range and gender

It can be seen that speakers under 30 are over-represented, which can be explained by the recruitment process in the University of Edinburgh. Apart from a few age ranges, female speakers tend also to be more represented than males. It is also interesting to analyze the repartition of regional accents depending on the age as shown in Figure 4.

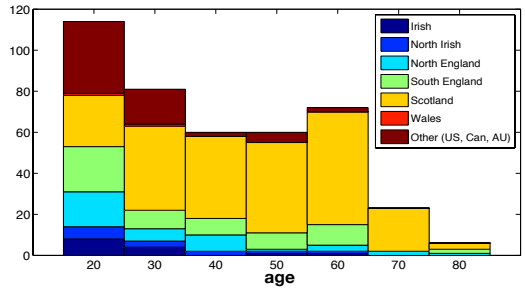


Figure 4: Distribution of speakers according to their age range and accent

As expected, there is a strong focus on scottish accent in overall but the balance between regional accents is also dependent to the age. Older speakers tend to be almost all scottish whereas the accent distribution is much more balanced for speakers under 25 which are mostly students. The latter is also confirmed by the repartition of regional accents depending on the social class as shown in Figure 5.

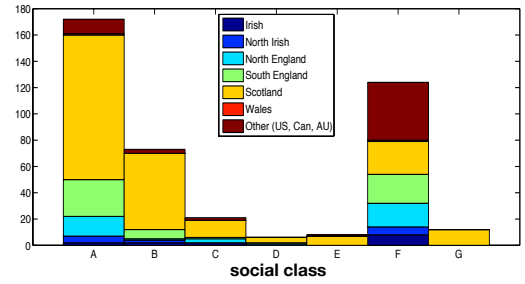


Figure 5: Distribution of speakers according to their accent and social classes as defined in section 3.3

The higher social classes are clearly over-represented whereas there is almost no "working class" speakers. This appears to be a drawback of our current recruitment procedure that reaches mostly higher classes and highly-educated profiles.

5. Conclusions

With more than 300 hours of speech data from approximately 500 healthy speakers, the Voice Bank corpus is already the largest speech database available in the UK. It has been designed to provide a large catalogue of voice donors for the voice reconstruction of individuals with speech disorders, and new recordings are continuously made to improve the coverage of various accents and voice identities. However, the recruitment of the speakers should be adapted in order to reach a larger variety of social classes. Another current limitation of the Voice Bank corpus is that it consists of "reading style" speech, which is not well suited for applications such as voice-output communication aids. Therefore, we are planning to record "conversational style" speech in the future. This could also allow a better representation of the regional accents since the accents are likely to be stronger in conversations that in formal, read speech. Nevertheless, we believe that the Voice Bank corpus is already a very useful resource for researcher in speech synthesis.

6. References

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